CSCI 39536 – Introduction to Robotics

Semester Robotics Project & Checkpoints

Weight: 30% of Final Grade
Team Project

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1 Overview

The semester project is the primary opportunity to integrate ideas from across the course into a coherent embodied robotic system. Your team will identify a meaningful problem or interaction, develop a robotic system that addresses it, evaluate how well the system works, and revise the design in response to technical evidence, stakeholder needs, and responsible-robotics considerations. The project is intentionally open-ended. You are **not** required to incorporate every technique taught in the course. Instead, your team should select and justify the sensing, perception, control, localization, planning, navigation, AI, or interaction methods that are appropriate for your problem. A focused system that makes thoughtful choices and is evaluated well is stronger than a complicated system containing disconnected features. The project develops through three checkpoints followed by a final demonstration and presentation.

2 Project Goals

By the end of the project, your team should be able to design and integrate an embodied robotic system; justify technical and human-centered design decisions; evaluate performance, uncertainty, limitations, and failure modes; implement responsible-design considerations where relevant; use AI-assisted development critically and transparently; and communicate the system clearly to others.

3 Project Requirements

Every project must involve an embodied robotic system operating in simulation or an approved physical environment. The system should sense or receive information about its environment, make decisions that affect its behavior, and perform a meaningful task or interaction. Projects should demonstrate meaningful technical integration using course concepts appropriate to the problem. Human-centered design must influence the project rather than appear only in the final presentation. Teams must identify relevant stakeholders, consider how people will interact with or be affected by the system, and evaluate at least one aspect of the system from a user or stakeholder perspective.

Responsible robotics must also influence implementation. Depending on the project, this may include safety constraints, accessible interaction, privacy protections, human override, uncertainty handling, transparency, bias mitigation, data minimization, or another justified design measure. Teams should be able to show what changed in the system because of these considerations. Evaluation is required throughout the project. Teams should define evidence that matters for their system rather than collect a fixed number of generic metrics. Evaluation should include technical performance and, where relevant, human-centered or responsible-design outcomes.

4 Project Evidence

The GitHub repository serves as the running technical record of the project. Teams should maintain source code, configuration files, documentation, setup or launch instructions, and other materials needed to understand and reproduce the system. At each checkpoint, teams will submit a GitHub commit link and a short checkpoint memo. The memo should be concise and may include diagrams, tables, figures, or links to demonstration evidence. Repeating material from earlier checkpoints is unnecessary; focus on what has changed, what you learned, and what decisions come next.

Significant use of generative AI should be disclosed throughout the project. Teams should identify how AI was used and how important outputs were checked, tested, or modified.

5 Checkpoint 1: Problem, Stakeholders & Initial Prototype

Due: October 15, 2026, before the in-class demonstrations
Value: 5% of Final Grade

Checkpoint 1 establishes the direction of the project. Your team should define the problem or opportunity, identify important users and stakeholders, explain the intended context of use, and demonstrate an initial working prototype or integrated subsystem. By this point, teams have encountered human-centered design, motion and kinematics, feedback control, AI-assisted programming, sensing, perception, localization, and mapping. Your prototype should use the subset of these ideas that is meaningful for your project; it does not need to demonstrate all of them.

The checkpoint memo should be no more than two pages and should explain the problem, stakeholders, initial requirements, current system architecture, what currently works, initial testing evidence, major limitations, and the next technical priorities. Include a system diagram and identify any major changes made because of stakeholder or contextual considerations. The in-class demonstration should show real system behavior rather than slides alone.

Checkpoint 1 Rubric

Category	Points	Criteria
Problem, Context, and Stakeholders	20	Defines a meaningful problem, relevant stakeholders, and appropriate project requirements.
Initial Technical Prototype	30	Demonstrates working robotic functionality appropriate to the project.
Architecture and Integration	20	Shows a coherent system structure and explains how components work together.
Testing and Learning	20	Provides evidence from testing, identifies limitations or failures, and uses results to guide next steps.
Communication and Documentation	10	Repository, memo, diagram, and demonstration clearly communicate current progress.

6 Checkpoint 2: Technical Integration & Autonomous Behavior

Due: November 5, 2026
Value: 5% of Final Grade

Checkpoint 2 demonstrates substantial technical progress beyond the initial prototype. The system should now integrate additional capabilities needed to perform its intended task more independently and reliably. Relevant course ideas may include computer vision or learned perception, localization and SLAM, planning, navigation, foundation models, sensor fusion, or other techniques appropriate to the project. These are options, not a checklist. Teams should be prepared to explain why they selected particular approaches and why other techniques were unnecessary or inappropriate.

The checkpoint memo should be no more than two pages and should update the system architecture, describe the major technical additions since Checkpoint 1, present meaningful performance evidence, analyze at least two important failure modes or limitations, and identify the next integration priorities. Where possible, include quantitative results rather than relying only on a successful demonstration.

Checkpoint 2 Rubric

Category	Points	Criteria
Technical Progress	30	Adds meaningful capabilities appropriate to the project and demonstrates clear development beyond Checkpoint 1.
System Integration	25	Components operate together as a coherent system rather than as disconnected demonstrations.
Performance and Reliability	20	Uses relevant evidence to evaluate whether the system performs its intended task reliably.
Failure Analysis and Iteration	15	Identifies important failure modes or limitations and explains how testing informs revision.
Communication and Documentation	10	Repository and memo clearly explain design choices, evidence, and next steps.

7 Checkpoint 3: Human-Centered, Responsible & Evaluation Review

Due: November 24, 2026
Value: 5% of Final Grade

Checkpoint 3 asks your team to evaluate the nearly complete system from technical, human-centered, and responsible-robotics perspectives and use that evaluation to guide final revisions. Teams should demonstrate how stakeholder needs or HRI considerations have influenced the design and how at least one responsible-design concern has been translated into system behavior, architecture, constraints, data practices, or interaction design. Discuss the issues most relevant to your project rather than attempting to address every possible ethical concern.

The checkpoint memo should be no more than three pages. It should summarize the current system, present technical and human-centered evaluation evidence, explain the responsible-design measures implemented, identify remaining risks or limitations, and specify the final revisions your team will make before presentation. If user testing or interaction evaluation is appropriate, include what you learned and how the design changed as a result.

Checkpoint 3 Rubric

Category	Points	Criteria
Human-Centered Design and HRI	25	Shows how stakeholder needs, interaction considerations, or user evidence influenced the system.
Responsible Design in Practice	25	Implements and justifies concrete measures addressing relevant safety, accessibility, privacy, fairness, transparency, human-control, or related concerns.
Evaluation Quality	25	Uses appropriate technical and human-centered evidence to assess performance and limitations.
Risk, Failure, and Revision Plan	15	Identifies remaining risks or failure modes and proposes justified final improvements.
Communication and Documentation	10	Repository and memo clearly communicate evidence, decisions, and final priorities.

8 Final Project Demonstration & Presentation

Due: Finals Week
Value: 15% of Final Grade

The final project should demonstrate a coherent system rather than a collection of course exercises. Teams should show the robot performing a meaningful task or interaction, explain the design choices that made that behavior possible, and present evidence about how well the system works and where it still fails.

The final submission consists of the completed GitHub repository, presentation slides and script, and a final project report of no more than seven pages. The report should synthesize rather than repeat the checkpoint memos. It should explain the problem and stakeholders, final system architecture and major design decisions, evaluation methods and results, important failures and limitations, human-centered and responsible-design choices, significant AI-assisted development and verification, and the most important future improvements.

Each team will give an approximately 10-minute presentation and live or recorded demonstration followed by questions. The presentation should emphasize the problem, the system as a whole, evidence from evaluation, important design tradeoffs, and what the team learned. All team members should participate and should be able to explain the overall system and major design decisions.

Final Project Rubric

Category	Points	Criteria
Technical Implementation	20	The system's chosen technical components are appropriate, functional, and well executed.
System Integration and Task Performance	20	The system operates coherently and meaningfully addresses the intended task or interaction.
Evaluation and Engineering Reasoning	20	Uses appropriate evidence to evaluate performance, uncertainty, tradeoffs, failure modes, and design decisions.

Human-Centered and Responsible Design	20	Stakeholder needs and relevant responsible-robotics considerations meaningfully influence the final system.
Communication and Demonstration	15	Presentation, demonstration, report, and responses to questions clearly communicate the project and its limitations.
AI-Assisted Development Transparency	5	Significant AI use is disclosed and important AI-generated contributions are appropriately verified or tested.

9 Scope and Expectations

Project ambition should be appropriate for the time and resources available. Reliability, thoughtful integration, evaluation, and justified design decisions matter more than the number of features. Using a library, pretrained model, ROS package, or existing algorithm is acceptable when appropriate; teams are expected to understand how it is being used, test its behavior, and explain its limitations.

Unexpected failures are part of robotics. A project does not need to work perfectly to be successful, but teams must test honestly, preserve evidence, explain what failed, and show how that evidence influenced their engineering decisions.

Individual contribution, collaboration, communication, and reflection are assessed separately through the *Teamwork, Collaboration & Individual Reflection* component of the course grade.